Radiation Detector Integration with Real-Time Data Recorder Application

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[IMAGE OF PROTOTYPE]

Technical Field

This project involves the technical fields of Embedded systems with integrated sensors, Application development, Data Management, Statistical Data Science, Kotlin, C/C++.

Background Information

Drawing from nearly a decade of operational experience and challenges in a naval nuclear environment, the project was conceived to address the critical shortcomings in current radiation survey methodologies. While modern monitoring equipment is effective and provides some mobility, it lacks ease of data logging and intuitive technological integration desired by field technicians, particularly in the US Naval Nuclear Propulsion Program. Inspired by these needs, this initiative combines embedded sensor technology with an offline-enabled tablet application to offer real-time, map-based data visualization and archival. This innovative approach aims to combine and enhance safety, accuracy, and efficiency of radiation surveys in environments where conventional, bulky devices are inadequate and where storage space for archived paper logs is limited.

Prior Art (Research)

Prior Art Disclaimer

There is prior art in the field of portable radiation detection devices. Although these existing solutions serve as valuable references for measuring radiation levels, they do not provide the cohesive offline mapping, integrated data logging, and archiving functionality proposed in this project. The following product illustrates how current technologies address core detection needs but leave gaps that are fulfilled by this innovation.

The RadEye PRD

A compact, highly sensitive personal radiation detector designed for quick deployment in various radiation monitoring scenarios. It offers audible and visual alarms and can store basic reading data.

Reference:

Thermo Fisher Scientific. (n.d.). *RadEye PRD personal radiation detector*. https://www.thermofisher.com/order/catalog/product/425069172

Project Description

This project focuses on creating a handheld radiation detection system for nuclear propulsion technicians and private-sector radiological technicians, typically 25–50 years old and tech-savvy, who often work in confined spaces requiring accurate, real-time measurements. A Geiger-Müller tube and Matrix Portal S3 microcontroller (programmed in C/C++) capture and log radiation data, which is transferred via a wired connection to a Kotlin-based Android tablet application. The tablet provides a secure, offline mapping interface where users can select locations, store sensor readings in a local SQLite database, and view real-time markers for immediate feedback. By streamlining data recording and auditing, the system aims to reduce administrative overhead and enhance operational efficiency in radiation surveys.

Innovation Claim

This solution is an improvement to existing designs because it provides a highly mobile, easy-to-use, and cost-effective alternative to bulky or proprietary systems. It enhances data accessibility for technicians conducting manual radiation surveys, ensuring accuracy and reliability.

Usage Scenario

In a naval nuclear power plant, the traditional reliance on paper logs for radiological surveys in confined spaces poses significant challenges, including limited physical storage, time-consuming data transfer, and the heightened risk of data loss or damage. The innovative handheld radiation detection system paired with an offline tablet application offers a transformative solution by digitally capturing real-time radiation readings and annotating them directly onto a secure facility map. This advanced approach eliminates the cumbersome manual process, dramatically reduces transcription errors, and ensures that safety-critical data is reliably archived in a local SQLite database. By streamlining administrative tasks and enhancing data integrity, the system not only improves operational efficiency but also significantly bolsters compliance with stringent safety protocols, ultimately providing a highly reliable tool for radiological maintenance in high-security environments.

SIP BRIEF PART 2

Evaluation Criteria

Does the Matrix Portal S3 successfully read the current generated by the Geiger-Muller tube and convert it to a measurable exposure rate?

Can the Android application generate a new map, and does it allow the user to save and load maps?

Is the current exposure rate reliably displayed on the Android application in real time?

Can the user log multiple types of surveys (radiation, airborne, contamination) onto the map?

Can the survey map be saved, restored with all survey data intact, and edited without loss of information?

Does the system operate entirely offline via a wired connection without the need for wireless communication?

Goals and Tasks Associated with the Project

Objective 1: Integrate Reliable Sensor Data Capture
Tasks:

- Develop C/C++ firmware on the Matrix Portal S3 to accurately read pulses from the Geiger-Müller tube.
- Implement an algorithm to convert pulse counts into a corresponding exposure rate.
- Establish and verify a stable, wired USB communication link to transmit data to the Android tablet.
- Validate sensor performance through controlled tests to confirm that the generated exposure rate is accurate and consistent.

Objective 2: Develop a Secure, Offline-Capable Android Application
Tasks:

- Design and implement a user-friendly interface that features an interactive map for selecting survey locations.
- Incorporate functionality for users to generate, save, load, and edit survey maps.
- Integrate a local SQLite database to store all sensor data and survey logs securely.

Description of Design Prototype

Describe the **design prototype** implementation. You should describe the platform on which the system will be built and provide directions on how to run the prototype (if necessary). Elaborate on the functionality of the parts that define your project. Effective descriptions will give the reader an understanding of what the design prototype will be, and how it relates to the final project.

<u>Note: This section will be revised prior to SIP408 to describe the design prototype in its final form.</u>

SIP Brief Part 3 – to be filled out in SIP409

Evaluation Plan

Provide a complete, paragraph-style description of the plan used to evaluate your project. This section should be a **description of the full plan** for how you will go about answering the "Evaluation Criteria" questions. Do not simply repeat the questions!

<u>Note: This section must be revised in SIP409 to describe the full evaluation plan as it was actually implemented.</u>

Project Completion Assessment

<u>Note: This section must be completed in SIP409</u>

Provide an in-depth description of the completion assessment of your project. Describe how well the completed components function and highlight the innovative facets of your design. This is sometimes known as a “Post-Mortem” or “Lessons-Learned Report”. A good approach for this section is to answer the following 4 questions: “What went right? What went wrong? What was learned throughout the process? What would be done differently if you had to do it again?

Appendices

<u>Note: This section must be completed in SIP408.</u>

Include as appendices any supporting material for this project, including charts, graphs, and other data; images associated with the project; or other documentation (e.g., a game design document or read-me file). Include any prior art that was used such as U.S. Patent Documents, Foreign Patent Documents, or other sources. Remember that this section should only be a list of additional files, not the actual data of the files!

Use the following format:

Appendix letter: description of item – file name

Example…

Appendix A: Game design document – myGameDoc.docx

Appendix B: 3D render of primary character – mainCharacter.jpg